

Integrated Information Theory through Markovian Dynamics

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OPINION

Integrated information theory: from consciousness to its physical substrate

Giulio Tononi, Melanie Boly, Marcello Massimini and Christof Koch

Abstract | In this Opinion article, we discuss how integrated information theory accounts for several aspects of the relationship between consciousness and the brain. Integrated information theory starts from the essential properties of phenomenal experience, from which it derives the requirements for the physical substrate of consciousness. It argues that the physical substrate of consciousness must be a maximum of intrinsic cause–effect power and provides a means to determine, in principle, the quality and quantity of experience. The theory leads to some counterintuitive predictions and can be used to develop new tools for assessing consciousness in non-communicative patients.

Consciousness is subjective experience — ‘what it is like’, for example, to perceive a scene, to endure pain, to entertain a thought or to reflect on the experience itself^{1–3}. When consciousness fades, as it does in dreamless sleep, from the intrinsic perspective of the experiencing subject, the entire world vanishes.

Consciousness depends on the integrity of certain brain regions and the particular content of an experience depends on the activity of neurons in parts of the cerebral cortex⁴. However, despite increasingly refined clinical and experimental studies, a proper understanding of the relationship between consciousness and the brain has yet to be established^{5,6}. For example, it is not known why the cortex supports consciousness when the cerebellum does not, despite having four times as many neurons^{7,8}, or why consciousness fades during deep sleep while the cerebral cortex remains active. There are also many other difficult questions about consciousness. Are patients with a functional island of cortex surrounded by widespread damage conscious, and if so, of what? Are newborn infants conscious? Are animals that display complex behaviours, but have brains very different from humans, conscious? Can intelligent machines be conscious?

To answer these questions, the empirical study of consciousness should be complemented by a theoretical approach. The reason why some neural mechanisms, but not others, should be associated with consciousness has been called ‘the hard problem’ because it seems to defy the possibility of a scientific explanation¹⁰. In this Opinion article, we provide an overview of the integrated information theory (IIT) of consciousness, which has been developed over the past few years^{1–3,11,12}. IIT addresses the hard problem in a new way. It does not start from the brain and ask how it could give rise to experience; instead, it starts from the essential phenomenal properties of experience, or axioms, and infers postulates about the characteristics that are required of its physical substrate. Moreover, IIT presents a mathematical framework for evaluating the quality and quantity of consciousness^{1–3,9}. We begin by providing a summary of the axioms and corresponding postulates of IIT and show how they can be used, in principle, to identify the physical substrate of consciousness (PSC). We then discuss how IIT explains in a parsimonious manner a variety of facts about the relationship between consciousness and

the brain, leads to testable predictions, and allows inferences and extrapolations about consciousness.

From phenomenology to physics

The axioms of IIT state that every experience exists intrinsically and is structured, specific, unitary and definite. IIT then postulates that, for each essential property of experience, there must be a corresponding causal property of the PSC. The postulates of IIT state that the PSC must have intrinsic cause–effect power; its parts must also have cause–effect power within the PSC and they must specify a cause–effect structure that is specific, unitary and definite. Below, we discuss the axioms and postulates of IIT (see [Supplementary information S1,S2](#) (figure, box)) and describe the fundamental identity — between an experience and a conceptual structure — that it proposes (FIG. 1).

The first axiom of IIT states that experience exists intrinsically. As recognized by Descartes¹³, my own experience is the only thing whose existence is immediately and absolutely evident, and it exists for myself, from my own intrinsic perspective. The corresponding postulate states that the PSC must also exist intrinsically. For something to exist in a physical sense, it must have cause–effect power — that is, it must be possible to make a difference to it (that is, change its state) and it must be able to make a difference to something. Moreover, the PSC must exist intrinsically — that is, it must have cause–effect power for itself, from its own intrinsic perspective. A neuron in the brain, for example, satisfies the criterion for existence because it has two or more internal states (such as active and inactive) that can be affected by inputs (causes) and its output can make a difference to other neurons (effects). A minimal system consisting of two interconnected neurons satisfies the criterion of intrinsic existence because, through their reciprocal interactions, the system can make a difference to itself.

The axiom of composition states that experience is structured, being composed of several phenomenal distinctions that exist within it. For example, within an experience, I may distinguish a piano, a blue colour, a book, countless spatial locations, and so on

RESEARCH ARTICLE

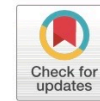
Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms

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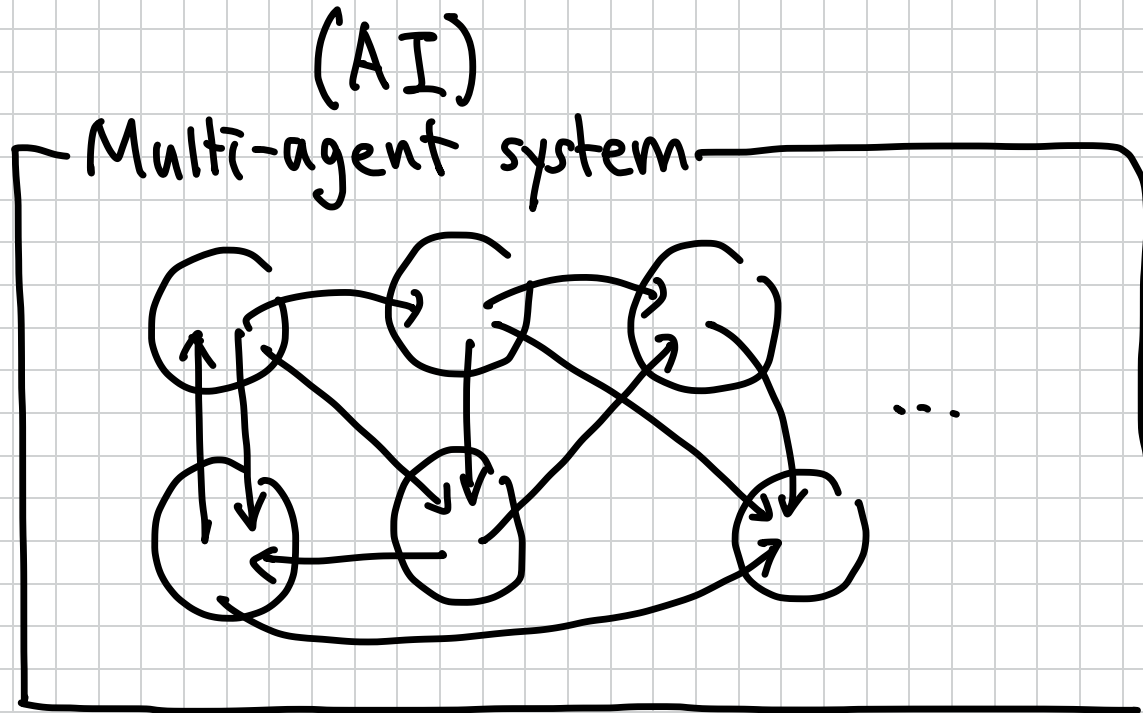
Abstract

This paper presents Integrated Information Theory (IIT) 4.0. IIT aims to account for the properties of experience in physical (operational) terms. It identifies the essential properties of experience (axioms), infers the necessary and sufficient properties that its substrate must satisfy (postulates), and expresses them in mathematical terms. In principle, the postulates can be applied to any system of units in a state to determine whether it is conscious, to what degree, and in what way. IIT offers a parsimonious explanation of empirical evidence, makes testable predictions concerning both the presence and the quality of experience, and permits inferences and extrapolations. IIT 4.0 incorporates several developments of the past ten years, including a more accurate formulation of the axioms as postulates and mathematical expressions, the introduction of a unique measure of intrinsic information that is consistent with the postulates, and an explicit assessment of causal relations. By fully unfolding a system’s irreducible cause–effect power, the distinctions and relations specified by a substrate can account for the quality of experience.

Author summary

As a theory of consciousness, IIT aims to answer two questions: 1) Why is experience present vs. absent? and 2) Why do specific experiences feel the way they do? The theory’s starting point is the existence of experience. IIT then aims to account for phenomenal existence and its essential properties in physical terms. It concludes that a substrate—a set of interacting units—can support consciousness if it can take and make a difference for itself (intrinsicity), select a specific cause and effect as an irreducible whole with a

어떻게 접하였나



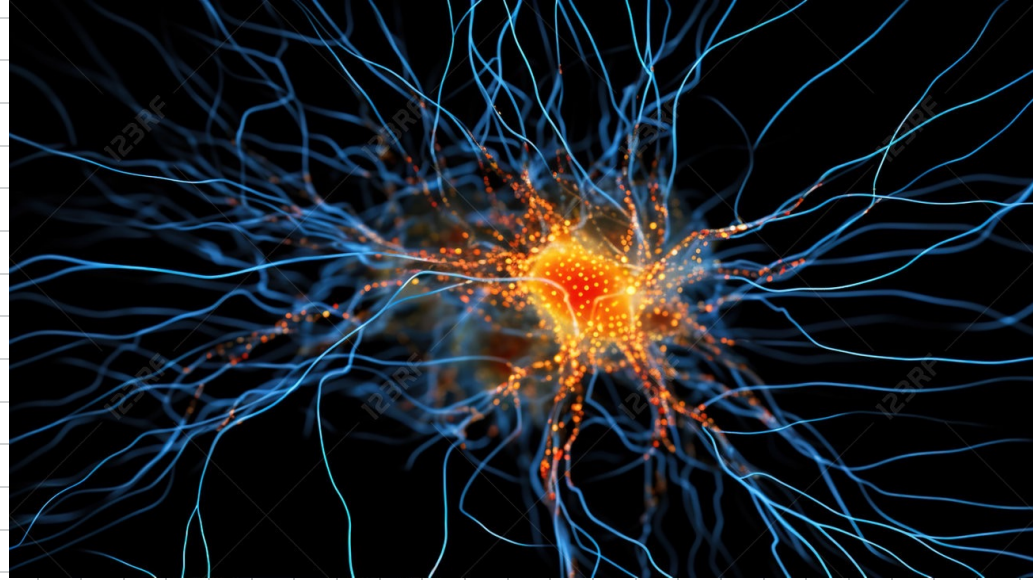
AI는 의식을 가질까?

= AI는 통합도가 사람의 뇌처럼 높을까?

AI가 의식을 가진다면, 다른 의식을 갖지 않는 AI보다 더 좋은 성능을 가질까?

... 2를 아주 짧게 살펴봤습니다.

"Relationship between consciousness and the brain"



의식 $\stackrel{?}{=}$ 뉴런의 신호(0,1)

우리는 어떻게 의식을 갖고 있는가?

Integrated Information Theory, IIT

Why do some physical systems support consciousness, while others do not?

- 의식 (consciousness)이란?

우리가 보고, 느끼고, 생각할 때 경험하는 subjective experience, "what it is like"

- 우리는 아직 의식과 뇌 사이의 관계를 모른다!

- 왜 어떤 뇌 영역은 의식과 관련되고, 어떤 영역은 그렇지 않은가?
- 의식은 인간에게만 있는가? 동물이나 인공지능도 의식을 가지고 있는가?

Integrated Information Theory, IIT

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- 의식은 인간에게만 있는가? 동물이나 인공지능도 의식을 가질 수 있는가?

⇒ IIT는 이 질문을 mathematical framework로 다루려는 이론.

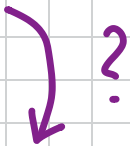
↳ Markovian Dynamics

의식에 대해 설명하는 다른 이론들과의 차별점?

It does not start from the brain and ask how it could give rise to experience; instead, it starts from the essential phenomenal properties of experience, or axioms, and infers postulates about the characteristics that are required of its physical substrate.

⇒ 뇌의 활동에서 의식을 설명하는 대신,
먼저 '경험이 가져야 할 성질'을 정의하고
이를 만족하는 물리적 조건을 수학적으로 정식화

Outline.

1. Axiom of experience
 2. Postulate of physical system
 3. Background about Markovian Dynamics
 4. Process that IIT shows
 - small ϕ φ
 - big ϕ Φ
- 

Axioms of IIT

Every experience exists

intrinsically and is structured, specific, unitary and definite.

내재성 (그 자체로 존재)

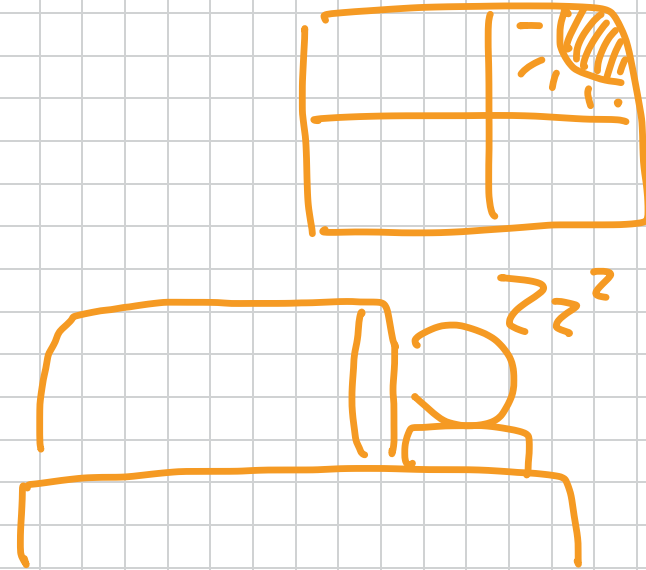
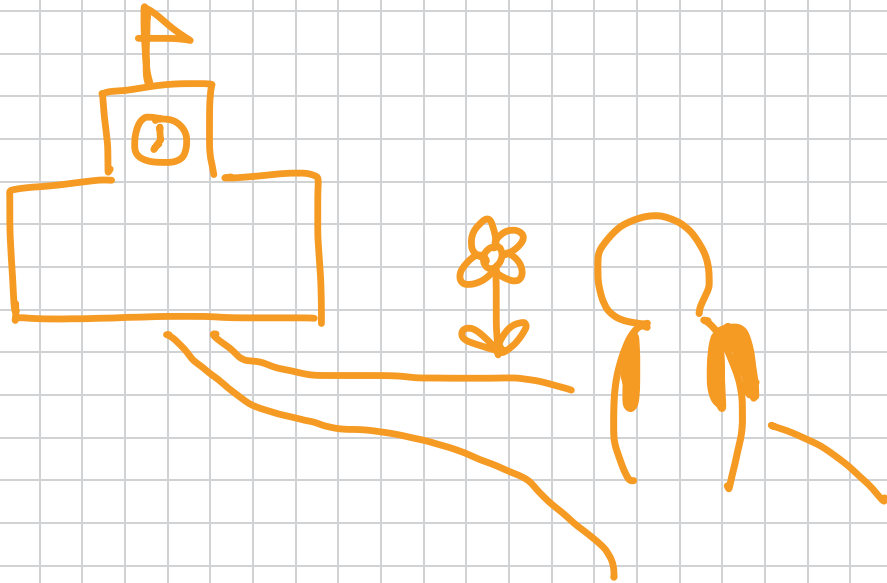
구성성 (구조화되어 있음)

정보성 (구체적임)

통합성 (단일함)

배제성 (명확함)

e.g.



Axioms of IIT

Every experience exists

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내재성 (그 자체로 존재)

구성성 (구조화되어 있음)

정보성 (구체적임)

통합성 (단일함)

배제성 (명확함)

나는 생각한다, 고로 존재한다
- 데카르트

Axioms of IIT

Every experience exists

intrinsically and is structured, specific, unitary and definite.

내재성 (그 자체로 존재)

구성성 (구조화되어 있음)

정보성 (구체적임)

통합성 (단일함)

배제성 (명확함)



Postulates of IIT

↑ Show how they can be used, in principle, to identify the physical substrate of consciousness (PSC).

의식(경험)이 물리적으로 가져야 할 속성, 특징은 무엇일까?

Postulates of IIT

The physical substrate must have *intrinsic* *cause-effect* power.

CAUSE-EFFECT STRUCTURE ★

Phenomenal property	Physical requirement
intrinsic	cause-effect power within itself
structured	subsets also have cause-effect power
specific	specifies a specific cause-effect state/structure
unitary	cause-effect structure is irreducible
definite	definite units/states/spatiotemporal gain

How can describe **cause-effect** mathematically?

i) How does the system change over time?

→ **Stochastic process!**

ii) If the next state depends only on the current state?

→ **Markov property!** (with transition probabilities)

This is the language IIT uses to describe system dynamics.

Background

Stochastic process

A sequence of random variables

$$X = \{X(t), t \in T\}$$

is called a stochastic process.

e.g. stock price prediction

Background

Markov property

A seq. $\{X_n\}_{n \in \mathbb{N}}$ of random variables possesses the Markov property, if for all $n \in \mathbb{N}$, it holds

$$P(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \dots, X_1 = x_1) = P(X_{n+1} = x_{n+1} \mid X_n = x_n)$$

e.g. Random walk

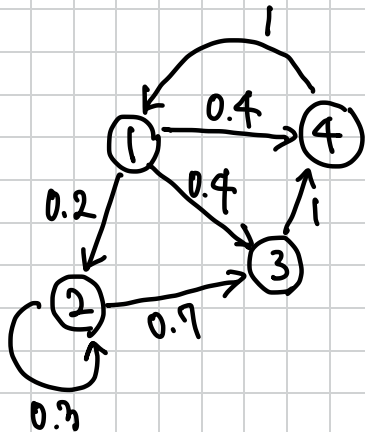
Background

Transition Probability Matrix (TPM)

Probability of moving from state i to state j in one time step is

$$P(j|i) = P_{ij}$$

e.g.



$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 0.2 & 0.4 & 0.4 \\ 0 & 0.3 & 0.7 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

"From Markovian Dynamics to a Complex"

i) Describe the system as Markovian dynamics.

시스템을 현재 상태에서 다음 상태로의 transition으로 표현

ii) Choose a candidate system.

전체 중 어느부분이 시스템 하나의 cause-effect system을 이루는지, 후보를 하나씩 관찰

iii) Look from its intrinsic perspective

후보 system 자체의 causal power를 알고 싶으면, 그 외의 요소는 background로 분리 후

iv) Trace cause and effect.

"어떤 과거 상태가 지금의 상태를 만들수 있었나?", "지금의 상태가 어떤 미래 상태를 만들 수 있는가?" 분석

v) Check whether it exists as one whole. Test irreducibility.

시스템을 여러부분으로 나누고, 부분 사이의 causal connection을 끊었을 때 원래의 cause-effect가 얼마나 약해지는지 확인

vi) Find the **complex**. 여러 후보 중 가장 irreducible 한 cause-effect power를 가지는 system 선택

i) Describe the system as Markovian dynamics.

Mathematical Setup

Let $U = \{U_1, U_2, \dots, U_n\}$ be a substrate of n interacting units.

For each unit U_i , let Ω_{U_i} denote its state space.

Then the state space of U is $\Omega_U = \prod_{i=1}^n \Omega_{U_i}$.

Let $u \in \Omega_U$ denote the current state, and $\bar{u} \in \Omega_U$ a possible state after one system update.

The complete transition probability function is

$$T_U \equiv p(\bar{u}|u), \quad u, \bar{u} \in \Omega_U$$

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The complete transition probability function is

$$T_U \equiv p(\bar{u}|u), \quad u, \bar{u} \in \Omega_U$$

→ 현재 state u 에서 한번 update 후 가능한 state $\bar{u}$ 로 전이할 확률
(TPM of size $|\Omega_U|$)

i) Describe the system as Markovian dynamics.

Assumptions on the dynamics

The system

- updates in discrete steps
- has a finite state space Ω_U
- its individual units (= random variables $U_i \in U$) are conditionally independent from each other given the preceding state of U :

$$p(\bar{u}|u) = \prod_{i=1}^n p(\bar{u}_i|u)$$

- can determine the conditional probabilities $\bar{p}$ for every system state, with

$$p(\bar{u}|u) = p(\bar{u} | \text{do}(u))$$

ii) Choose a candidate system.

e.g. 한 학교의 교실을 보보자.

그중 학생 A, B를 골랐을 때 ($S = \{A, B\}$)

이 둘이 주변 학생들, 환경과 구분했을 때 단들이 얼마나 될까?

$$\begin{cases} U = \{A, B, C, \dots, Z\} : \text{전체 교실} \\ S = \{A, B\} : \text{분석할 candidate system} \\ W = U \setminus S : \text{candidate system 바깥의 학생, 환경들} \\ w = \text{바깥 학생, 환경들의 현재 상태} \end{cases}$$

$\Rightarrow$ A, B 사이의 관계와 행동은 주변의 영향을 받을 수 있다.

따라서 주변의 현재 상태 w 는 background로 두고, 그 조건 아래에서 A, B 들 자체의 cause-effect relation을 살펴본다!

ii) Choose a candidate system.

Starting from a substrate U in current state u ,

$S \subseteq U$ is a candidate system, and $s \subseteq u$ is its current state.

(S and u are sets of tuples containing both the units and their states)

By the intrinsic postulate, $W = U \setminus S$ are treated as background conditions and do not contribute directly to the cause-effect power of the system.

($w = u \setminus s$: current state of the background units.)

iii) Look from its intrinsic perspective.

To discount the contribution of background units, they are causally marginalized.

- background units are marginalized based on a uniform marginal distribution, updated by conditioning on u .
- this is done separately for each unit in the system
- the results are combined using a product, eliminating residual correlations due to the background units

From this intrinsic perspective, the cause-effect power of S is evaluated in both directions.

possible cause $\rightarrow S \rightarrow$ possible effect

- **Effect**: what the current state S can produce
- **Cause**: what could have produced the current state S

iii) Look from its intrinsic perspective.

Accordingly, we obtain two TPMs T_e and T_c , for evaluating effects and causes, respectively.

Effect TPM

For evaluating effects,

the state of the background units is fully determined by the current state of the universe.

$$T_e = T_e(T_u, u, w) \equiv p_e(\bar{s}|s) = p(\bar{s}|s, w), \quad s, \bar{s} \in \Omega_s$$

where $w = u \setminus s$.

iii) Look from its intrinsic perspective.

Accordingly, we obtain two TPMs T_e and T_c , for evaluating effects and causes, respectively.

Cause TPM

For evaluating causes,

knowledge of the current state is used to compute the probability distribution over potential prior states of the background units, which is not necessarily uniform or deterministic.

$$T_c = T_c(T_u, u, w) \equiv p_c(s|\bar{s}) = \frac{|s|}{\prod_{i=1}^{|s|}} \sum_{\bar{w}} p(s_i|\bar{s}, \bar{w}) \left(\frac{\prod_{\hat{s}} p(u|\hat{s}, \bar{w})}{\prod_{\hat{u}} p(u|\hat{u})} \right), \quad s, \bar{s} \in \Omega_s$$

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possible prior background state $\bar{w}$ 를 현재 u 에 따라 weighting

$$\hookrightarrow p(\bar{w}|u)$$

$$p(\bar{w}|u) = \frac{p(u|\bar{w})p(\bar{w})}{p(u)}$$

iv) Trace cause and effect.

e. g. 다시 학생 A, B를 보자.

현재 A와 B가 함께 문제를 풀고 있다고 하자.

이 상태는 이후 두 학생의 행동에 실제로 얼마나 큰 차이를 만들까?

$$\begin{cases} \mathcal{S} = \{A, B\} \\ S = \{A \text{와 } B \text{가 함께 문제를 풀}\} \end{cases}$$

$\bar{S} = \{A \text{와 } B \text{가 함께 공부함}\}$ 에 대해 $P_e(\bar{S}|S) = 0.8$ 을 얻었다고 하자.

즉, 현재 S일때 $\bar{S}$ 가 될 확률이 0.8이다.

iv) Trace cause and effect.

e. g. 다시 학생 A, B를 보자.

현재 A와 B가 함께 문제를 풀고 있다고 하자.

이 상태는 이후 두 학생의 행동에 실제로 얼마나 큰 차이를 만들까?

$$P_e(\bar{S}|s) = 0.8$$

i) $P_e(\bar{S}) = 0.75$ 이면

A와 B가 함께 문제를 풀고 있지 않더라도 $\bar{S}$ 가 비슷한 정도로 자주 나타날

→ 현재 상태가 큰 차이를 만들었다고 보기 어려움

ii) $P_e(\bar{S}) = 0.1$ 이면

현재는 $\bar{S}$ 가 잘 나타나지 않는데 현재 상태에 대해서는 크게 높아짐

→ S 가 $\bar{S}$ 가 실제로 강하게 만들어냄!

iv) Trace cause and effect.

e. g. 자식 학생 A, B를 보자.

현재 A와 B가 함께 문제를 풀고 있다고 하자.

이 상태는 이후 두 학생의 행동에 실제로 얼마나 큰 차이를 만들까?

$$P_e(\bar{S}|S) = 0.8$$

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→ S 가 $\bar{S}$ 가 실제로 강하게 만들어짐!

⇒ "현재 상태 S 가
자기 자신에 대해
특정한 cause/effect state를
얼마나 강하게 지정하는지"
정량화 해보자! 라는 것이
intrinsic information (ii)

iv) Trace cause and effect.

(Given a causal model with corresponding TPMs T_e and T_c , identify the maximal cause-effect state specified by the system in its current state over itself, and quantify the causal power with which it does so.

System intrinsic information ii

$$ii(s, \bar{s})$$

measures the causal power of a system S over itself, for its current state s , over a specific possible cause or effect state $\bar{s}$.

iv) Trace cause and effect.

Intrinsic information depends on interventional conditional probabilities and unconstrained probabilities of cause or effect states and is the product of selectivity and informativeness.

Intrinsic effect information $\bar{i}_e$

$$\bar{i}_e(s, \bar{s}) = p_e(\bar{s}|s) \log \left(\frac{p_e(\bar{s}|s)}{p_e(\bar{s})} \right)$$

$$\text{with } p_e(\bar{s}) = |\Omega_s|^{-1} \sum_{s \in \Omega_s} p_e(\bar{s}|s)$$

iv) Trace cause and effect.

Intrinsic information depends on interventional conditional probabilities and unconstrained probabilities of cause or effect states and is the product of selectivity and informativeness.

Intrinsic effect information $\bar{i}i_e$

selectivity: causal power가 여러 가능한 state에 퍼지지 않고 특정 cause/effect state에 얼마나 집중되어 있는가
informativeness: 특정 cause/effect state의 prob.가 chance (unconstrained prob.)에 비해 얼마나 증가했는가

$$\bar{i}i_e(s, \bar{s}) = p_e(\bar{s}|s) \log \left(\frac{p_e(\bar{s}|s)}{p_e(\bar{s})} \right)$$

$$\text{with } p_e(\bar{s}) = |\Omega_s|^{-1} \sum_{s \in \Omega_s} p_e(\bar{s}|s)$$

iv) Trace cause and effect.

Intrinsic cause information $\hat{i}i_c$

$$\hat{i}i_c(s, \bar{s}) = p_c^{\leftarrow}(\bar{s}|s) \log \left(\frac{p_c(s|\bar{s})}{p_c(s)} \right)$$

$$\text{with } p_c(s) = |\Omega_s|^{-1} \sum_{\bar{s} \in \Omega_s} p_c(s|\bar{s}),$$

$$p_c^{\leftarrow}(\bar{s}|s) = \frac{p_c(s|\bar{s}) \cdot |\Omega_s|^{-1}}{p_c(s)} = \frac{p_c(s|\bar{s})}{\sum_{\hat{s} \in \Omega_s} p_c(s|\hat{s})}$$

iv) Trace cause and effect.

Intrinsic cause information $\hat{i}i_c$

forward prob.: can $\bar{s}$ produce s ?

$$\hat{i}i_c(s, \bar{s}) = p_c^{\leftarrow}(\bar{s}|s) \log \left(\frac{p_c(s|\bar{s})}{p_c(s)} \right)$$

backward prob.: does s specifically select $\bar{s}$?

$$\text{with } p_c(s) = |\Omega_s|^{-1} \sum_{\bar{s} \in \Omega_s} p_c(s|\bar{s}),$$

$$p_c^{\leftarrow}(\bar{s}|s) = \frac{p_c(s|\bar{s}) \cdot |\Omega_s|^{-1}}{p_c(s)} = \frac{p_c(s|\bar{s})}{\sum_{s \in \Omega_s} p_c(s|\hat{s})}$$

iv) Trace cause and effect.

Intrinsic cause information $\hat{i}i_c$

$$\hat{i}i_c(s, \bar{s}) = p_c^{\leftarrow}(\bar{s}|s) \log \left(\frac{p_c(s|\bar{s})}{p_c(s)} \right)$$

$$\text{with } p_c(s) = |\Omega_s|^{-1} \sum_{\bar{s} \in \Omega_s} p_c(s|\bar{s}),$$

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$$p_c^{\leftarrow}(\bar{s}|s) = \frac{p_c(s|\bar{s}) \underbrace{p(\bar{s})}_{\frac{1}{|\Omega_s|} : \text{uniform prior}}}{p_c(s)}$$

iv) Trace cause and effect.

Intrinsic cause information $\bar{i}i_c$

selectivity & informativeness

$$\bar{i}i_c(s, \bar{s}) = p_c^{\leftarrow}(\bar{s}|s) \log \left(\frac{p_c(s|\bar{s})}{p_c(s)} \right)$$

$$\text{with } p_c(s) = |\Omega_s|^{-1} \sum_{\bar{s} \in \Omega_s} p_c(s|\bar{s}),$$

$$p_c^{\leftarrow}(\bar{s}|s) = \frac{p_c(s|\bar{s}) \cdot |\Omega_s|^{-1}}{p_c(s)} = \frac{p_c(s|\bar{s})}{\sum_{\bar{s} \in \Omega_s} p_c(s|\bar{s})}$$

iv) Trace cause and effect.

The maximal cause-effect state

$$\begin{aligned} s_e'(T_e, s) &= \operatorname{argmax}_{\bar{s} \in \Omega_s} iie(s, \bar{s}) \\ &= \operatorname{argmax}_{\bar{s} \in \Omega_s} p_e(\bar{s}, s) \log \left(\frac{p_e(\bar{s}, s)}{p_e(\bar{s})} \right) \end{aligned}$$

$$\bar{i}ie(T_e, s) := \bar{i}ie(s, s_e') = \max_{\bar{s} \in \Omega_s} p_e(\bar{s} | s) \log \left(\frac{p_e(\bar{s} | s)}{p_e(\bar{s})} \right)$$

The system selects a specific cause or effect state.

e.g. $\bar{i}ie(s, \bar{s}_1) = 0.2$, $\bar{i}ie(s, \bar{s}_2) = 1.1$, $\bar{i}ie(s, \bar{s}_3) = 0.4$ 이때
 $\bar{s}_2$ 가 현재 s 에 의해 가장 강하게 specified 되는 effect state

iv) Trace cause and effect.

The maximal cause-effect state

$$\begin{aligned} s_e'(T_e, s) &= \operatorname{argmax}_{\bar{s} \in \Omega_s} iie(s, \bar{s}) \\ &= \operatorname{argmax}_{\bar{s} \in \Omega_s} p_e(\bar{s}, s) \log \left(\frac{p_e(\bar{s}, s)}{p_e(\bar{s})} \right) \end{aligned}$$

$$\Rightarrow \boxed{s' = \{s_c', s_e'\}}$$

$$\bar{i}i_e(T_e, s) := \bar{i}i_e(s, s_e') = \max_{\bar{s} \in \Omega_s} p_e(\bar{s} | s) \log \left(\frac{p_e(\bar{s} | s)}{p_e(\bar{s})} \right)$$

The system selects a specific cause or effect state.

e.g. $\bar{i}i_e(s, \bar{s}_1) = 0.2$, $\bar{i}i_e(s, \bar{s}_2) = 1.1$, $\bar{i}i_e(s, \bar{s}_3) = 0.4$ 2021
 $\bar{s}_2$ 가 현재 s 에 의해 가장 강하게 specified 되는 effect state

v) Check whether it exists as one whole. Test irreducibility.

- maximal cause-effect state $s' = \{s_c', s_e'\}$ of a candidate system S in its current state s

⇒ Let's evaluate whether the system is irreducible!

(integration postulate: System cannot be subdivided into subsets of units that exist separately from one another.)

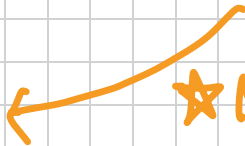
e.g. 다시 학생 A, B를 보자. 각 학생의 state는 1 = 문제풀이 이해함, 0 = 이해하지 못함이라고 하자.

$S = \{A, B\}$, $S = (1, 0)$ 일 때, S 가 가장 강하게 지칭하는 maximal effect state $s_e' = (1, 1)$ 이라고 하자.

그러면, 이 effect은 정말 A, B가 하나의 시스템으로 상호작용하기 때문에 나타나는 걸까?

→ 부분 사이의 연결을 끊어도 아무 차이가 없다면, 그 cause-effect power는 그 system만의 것이 아니다!

v) Check whether it exists as one whole. Test irreducibility.

Directional system partitions  ★ Main idea.

A set of directional system partitions $\Theta(S)$ divide S into $k \geq 2$ parts $\{S^{(i)}\}_{i=1}^k$, s.t.

$$S^{(i)} \neq \emptyset, S^{(i)} \cap S^{(j)} = \emptyset, \text{ and } \bigcup_{i=1}^k S^{(i)} = S$$

A partition $\theta \in \Theta(S)$ thus has the form

$$\theta = \{S_{\delta_1}^{(1)}, S_{\delta_2}^{(2)}, \dots, S_{\delta_k}^{(k)}\}$$

where $\delta_i \in \{\leftarrow, \rightarrow, \leftrightarrow\}$ indicates

whether inputs ($\leftarrow$), outputs ($\rightarrow$), or both ($\leftrightarrow$) are cut for a given part.

v) Check whether it exists as one whole. Test irreducibility.

$$\theta = \{S_{\delta_1}^{(1)}, S_{\delta_2}^{(2)}, \dots, S_{\delta_k}^{(k)}\}, \quad \delta_i \in \{\leftarrow, \rightarrow, \leftrightarrow\}$$

For each $S^{(i)}$,

$X^{(i)} \subseteq S$ is a set of units whose **inputs** to $S^{(i)}$ have been **cut by partition**

$$X^{(i)} = \begin{cases} S \setminus S^{(i)} & \text{if } \delta_i \in \{\leftarrow, \leftrightarrow\} \\ \bigcup_{\substack{j \neq i, \\ \delta_j \in \{\rightarrow, \leftrightarrow\}}} S^{(j)} & \text{if } \delta_i \in \{\rightarrow\} \end{cases}$$

v) Check whether it exists as one whole. Test irreducibility.

Partitioned transition probability matrix. T_e^θ, T_c^θ

Given a partition $\theta \in \Theta(s)$,

$$T_e^\theta \equiv p_e^\theta(\bar{s}, s) = \prod_{j=1}^n p_e^\theta(\bar{s}_j | s), \quad \bar{s}, s \in \Omega_s$$

(conditional independence)

where the partitioned probability of a unit $s_j \in \mathcal{S}^{(i)}$ is defined as

$$p_e^\theta(\bar{s}_j | s) = |\Omega_{x^{(i)}}|^{-1} \sum_{x^{(i)} \in \Omega_{x^{(i)}}} p_e(\bar{s}_j | x^{(i)}, y^{(i)}), \quad y^{(i)} = s \setminus x^{(i)}$$

All connections to unit s_j that are affected by the partition are causally marginalized.
(replaced by independent noise)

v) Check whether it exists as one whole. Test irreducibility.

System integrated information

The integrated effect information φ_e measures how much the partition $\theta \in \Theta_S$ reduces the probability with which a system S in state $s \in \Omega_S$ specifies its effect state se' ,

$$\varphi_e(T_e, s, \theta) = p_e(se'|s) \left| \log \left(\frac{p_e^{original}(se'|s)}{p_e^{\theta}(se'|s)} \right) \right|_+ \rightarrow \text{positive part operator}$$

Simillary,

$$\varphi_c(T_c, s, \theta) = p_c^{\leftarrow}(s_c'|s) \left| \log \left(\frac{p_c(s|s_c')}{p_c^{\theta}(s|s_c')} \right) \right|_+$$

v) Check whether it exists as one whole. Test irreducibility.

System integrated information

Then the system integrated information for a given partition is the minimum of its irreducibility on the cause and effect sides:

$$\varphi_s(T_e, T_c, s, \theta) = \min \{ \varphi_c(T_c, s, \theta), \varphi_e(T_e, s, \theta) \}$$

v) Check whether it exists as one whole. Test irreducibility.

Minimum partition (MIP)

By the principle of minimal existence, the integrated information of a system is given by its irreducibility over its minimum partition (MIP) $\theta' \in \Theta_s$, s.t

$$\varphi_s(T_e, T_c, s) := \varphi_s(T_e, T_c, s, \theta')$$

where

$$\theta' = \underset{\theta \in \Theta(s)}{\operatorname{argmin}} \frac{\varphi_s(T_e, T_s, s, \theta)}{\max_{T_e', T_c'} \varphi_s(T_e', T_c', s, \theta)}$$

v) Check whether it exists as one whole. Test irreducibility.

Minimum partition (MIP) → 어디를 끊으면 system의 cause-effect power를 가장 적게 손상시키면서 나눌수 있는가.

By the principle of minimal existence, the integrated information of a system is given by its irreducibility over its minimum partition (MIP) $\theta' \in \Theta_s$, s.t

$$\varphi_s(T_e, T_c, s) := \psi_s(T_e, T_c, s, \theta')$$

Accordingly, the system is reducible if at least one partition $\theta \in \Theta_s$ makes no difference to the cause or effect probability. Main idea★

φ (small phi) : quantifies the extent to which system S in state s has cause-effect power over itself as one system.

vi) Find the complex.

지금까지, 각 candidate system에 대해

φ_s = how irreducibly it exists as one system

를 구했다.

하지만 하나의 substrate 안에는 서로 overlap 하는 여러 candidate systems 가 있을수 있고,
각각 $\varphi_s > 0$ 인수도 있다.

그러면, 어떤 system이 하나의 definite system으로 존재할까?

vi) Find the complex.

Exclusion postulate에 따라, 서로 경쟁하는 candidate 중 system integrated information ϕ_S 가 가장 큰 candidate를 선택한다.

maximum $\phi_S \Rightarrow$ maximal substrate = complex

그리고 그 complex와 overlap 하면서 더 작은 ϕ_S 를 가지는 candidate system은 제외된다.

(principle of maximal existence) "the one that exists is the one that exists the most"

Beyond the complex.

Unfolding the Φ structure.

→ Complex의 내부 cause-effect power가 어떻게 구현되어있는가

$$\Phi = \sum_{\text{distinctions, relations}} \phi$$